

The Language Alphabet Σ

letters (a-z, A-Z)

digits (0-9)

operators (+, -, *, /, =, <, :, ;, (,), ,)

delimiters ({, }, "))

whitespace (space, tab, newline)

The Regular Expressions (RE) for all the lexical components of the TINY language

$letter(letter \mid digit)^* \mid digit^+ \mid (if \mid then \mid else \mid end \mid repeat \mid until \mid read \mid write) \mid "[any]^*" \mid "$

$\mid (+ \mid - \mid * \mid / \mid = \mid < \mid :=) \mid (; \mid , \mid (\mid)) \mid \{[any]^*\} \mid (space \mid tab \mid newline)^+$

1. Identifiers

An identifier consists of a letter followed by any number of letters or digits.

$letter(letter \mid digit)^*$

2. Numbers

A number is defined as a sequence of digits.

$digit^+$

3. Reserved Words (Keywords)

The TINY language has a specific set of reserved keywords.

if | then | else | end | repeat | until | read | write

4. Strings

Strings are enclosed in double quotes.

*"[any] *"*

5. Operators and Special Symbols

Operators

*+ | - | * | / | = | < | :=*

Punctuation

; | , | (|)

6. Whitespace and Comments

Comments

Comments are enclosed in curly brackets `{ }` and cannot be nested.

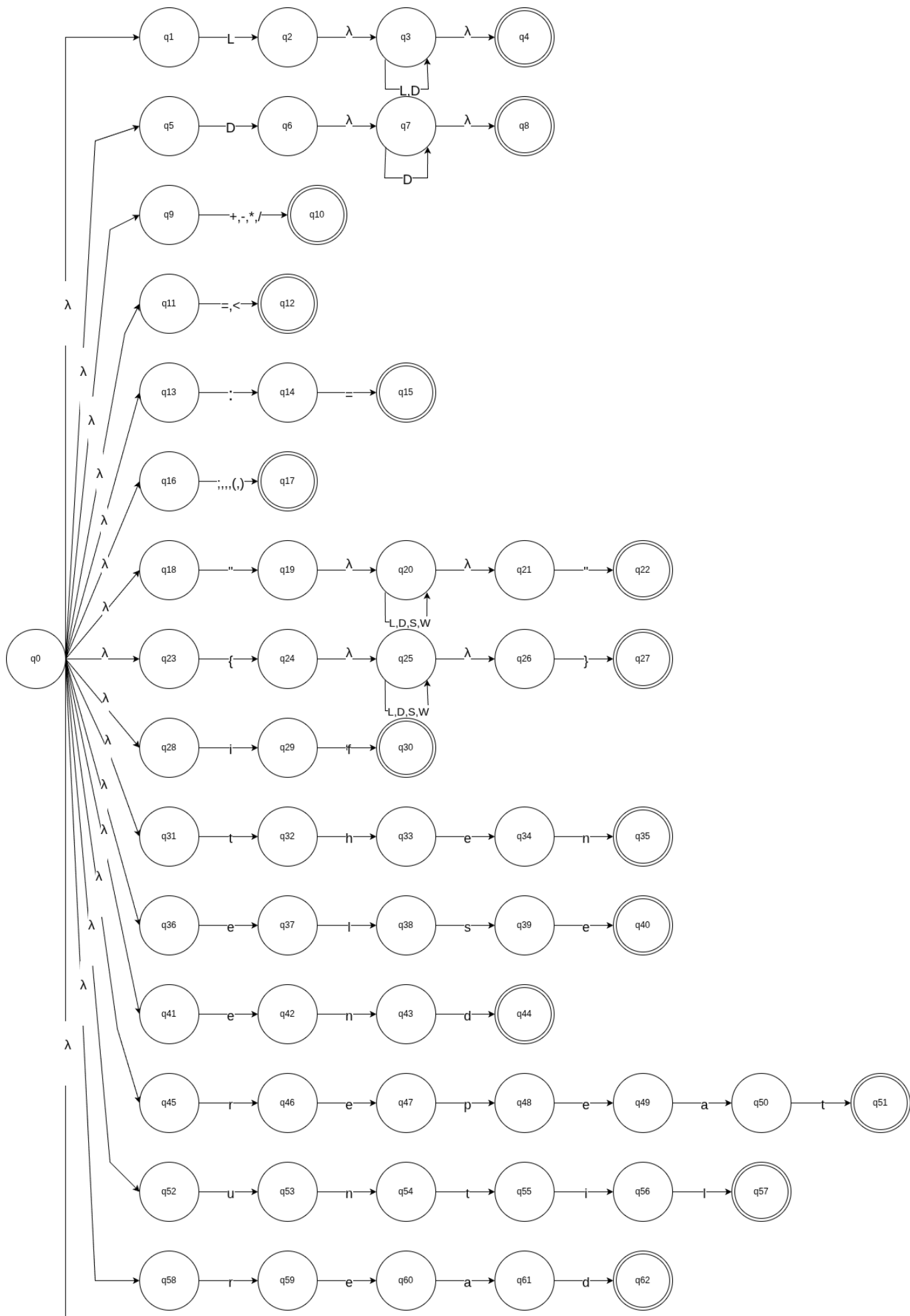
{[any]}*

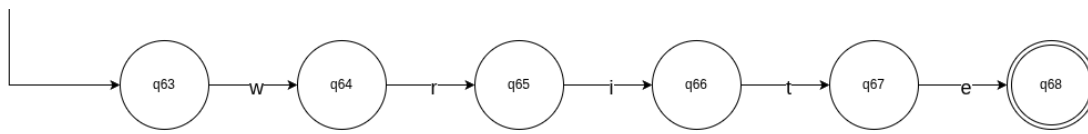
Whitespace

Whitespace includes spaces, tabs, and newlines.

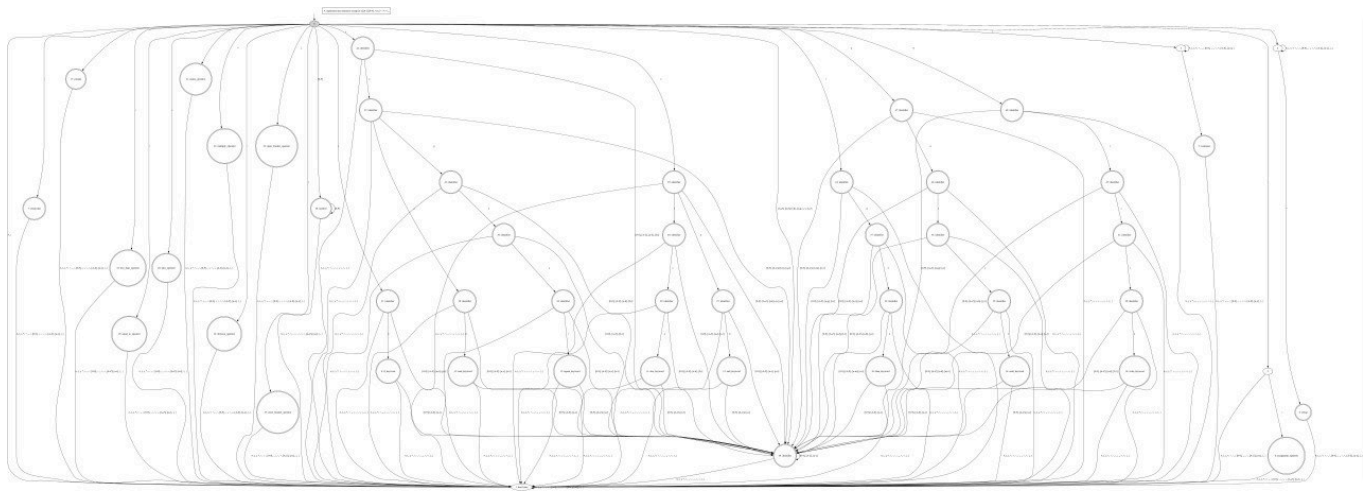
$(space \mid tab \mid newline)^+$

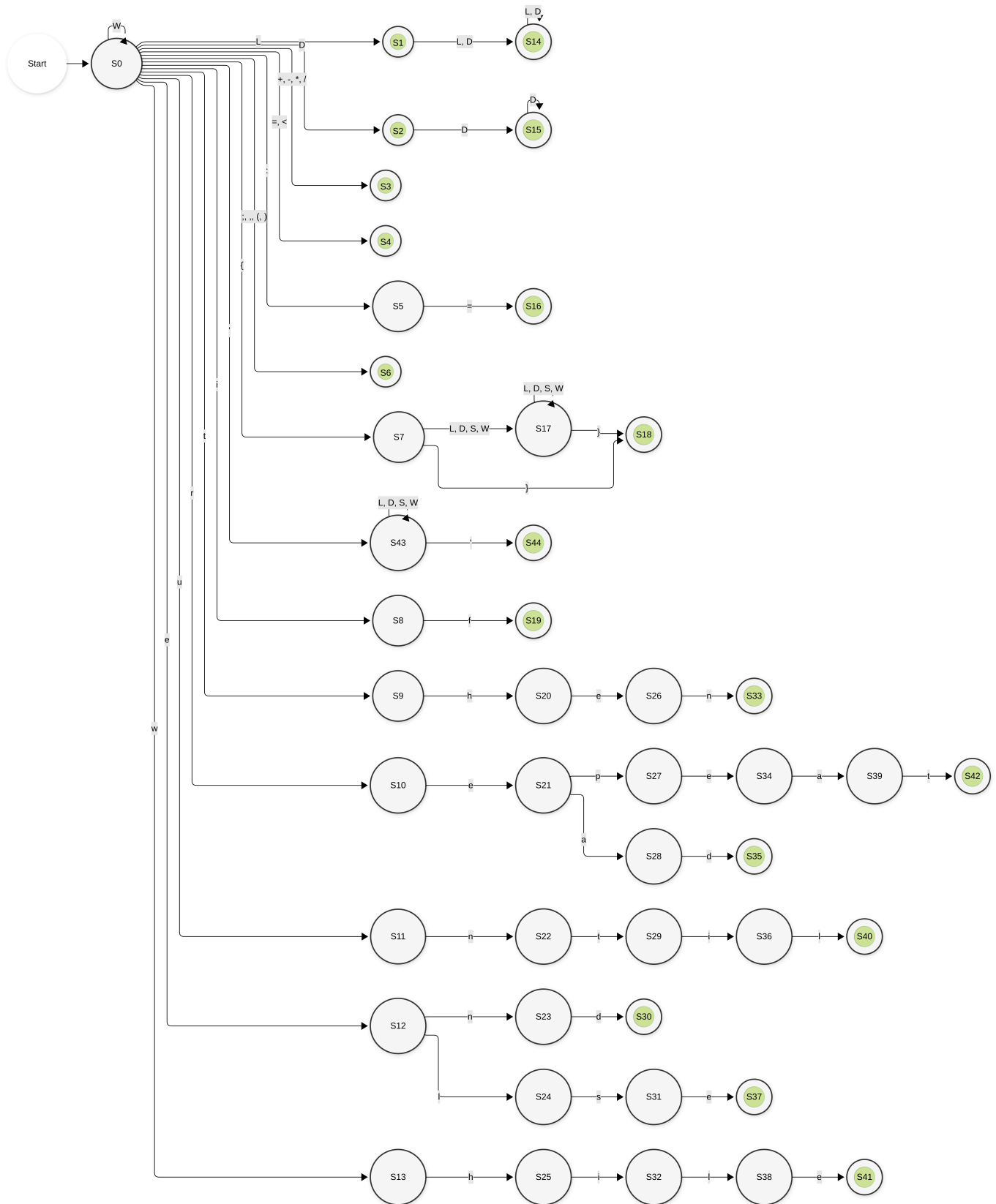
TINY Language - NFA Design





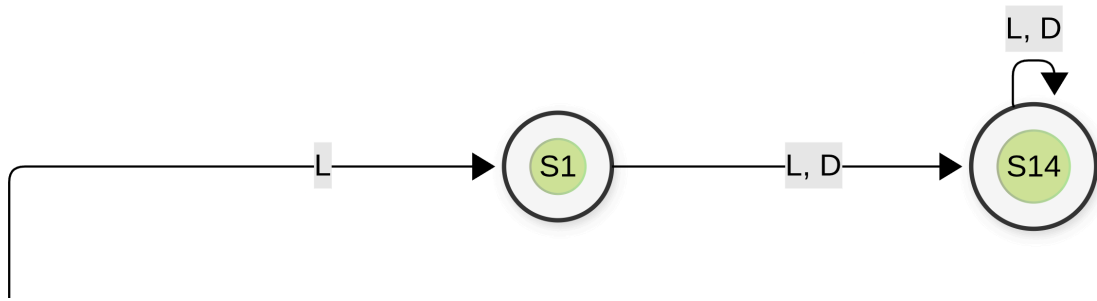
TINY Language - DFA Design





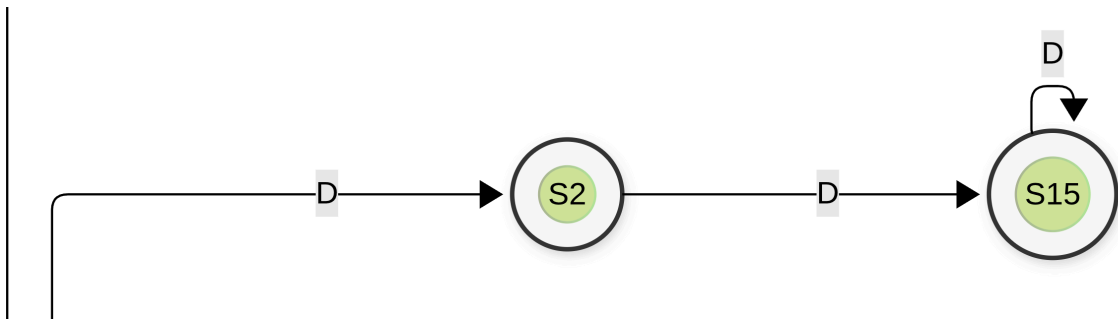
1. Identifiers

An identifier starts with a letter followed by letters or digits.


$$(q0) \xrightarrow{\text{letter}} (q1)$$
$$(q1) \xrightarrow{\text{letter/digit}} (q14)$$

- ✓ Accepting state: q14
- ✓ If lexeme matches keyword → Keyword
- ✓ Otherwise → Identifier

2. Numbers

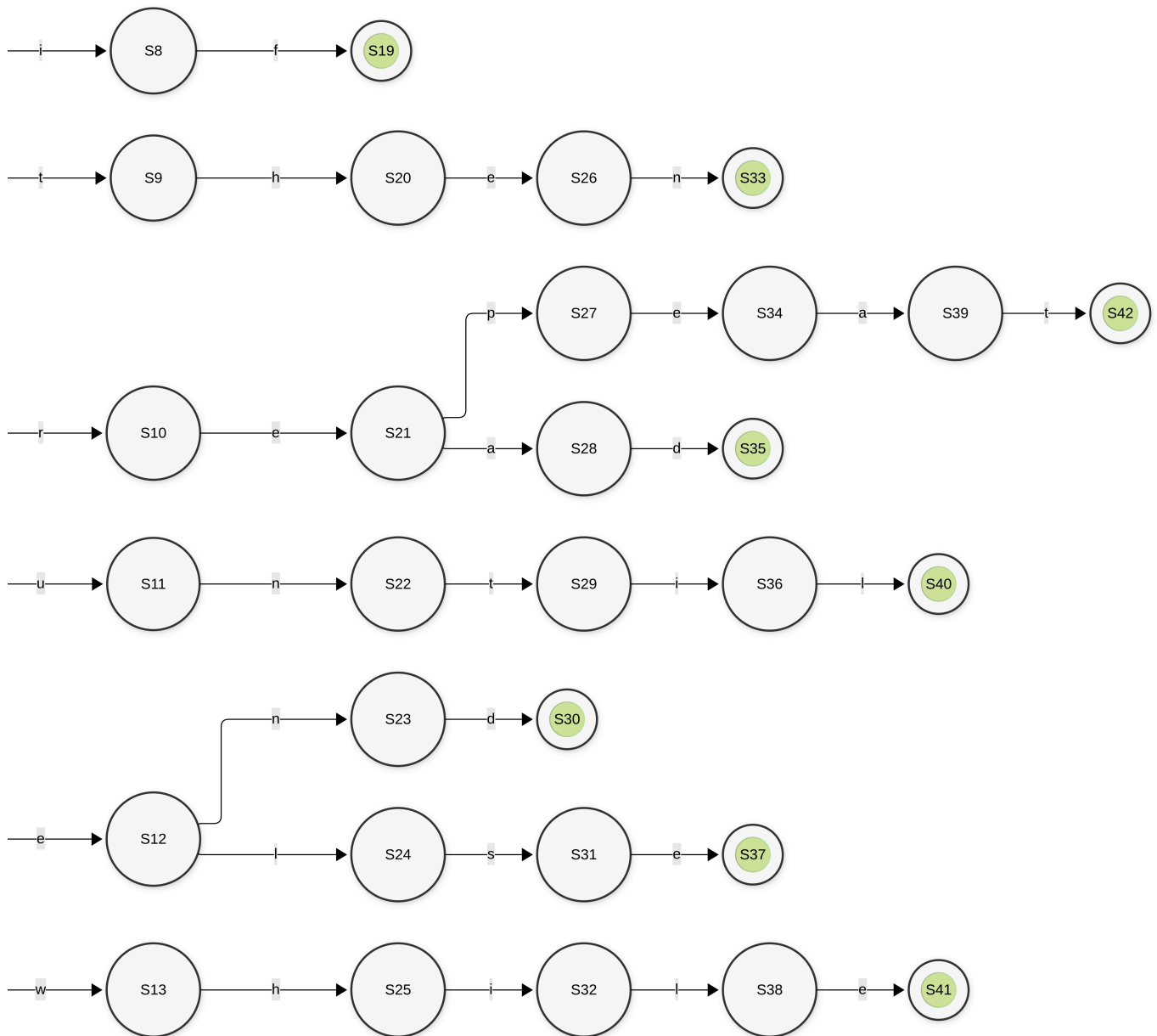


A number is a sequence of digits.

$$(q0) \xrightarrow{\text{digit}} (q2)$$
$$(q2) \xrightarrow{\text{digit}} (q15)$$

- ✓ Accepting state: q15

3. Keywords



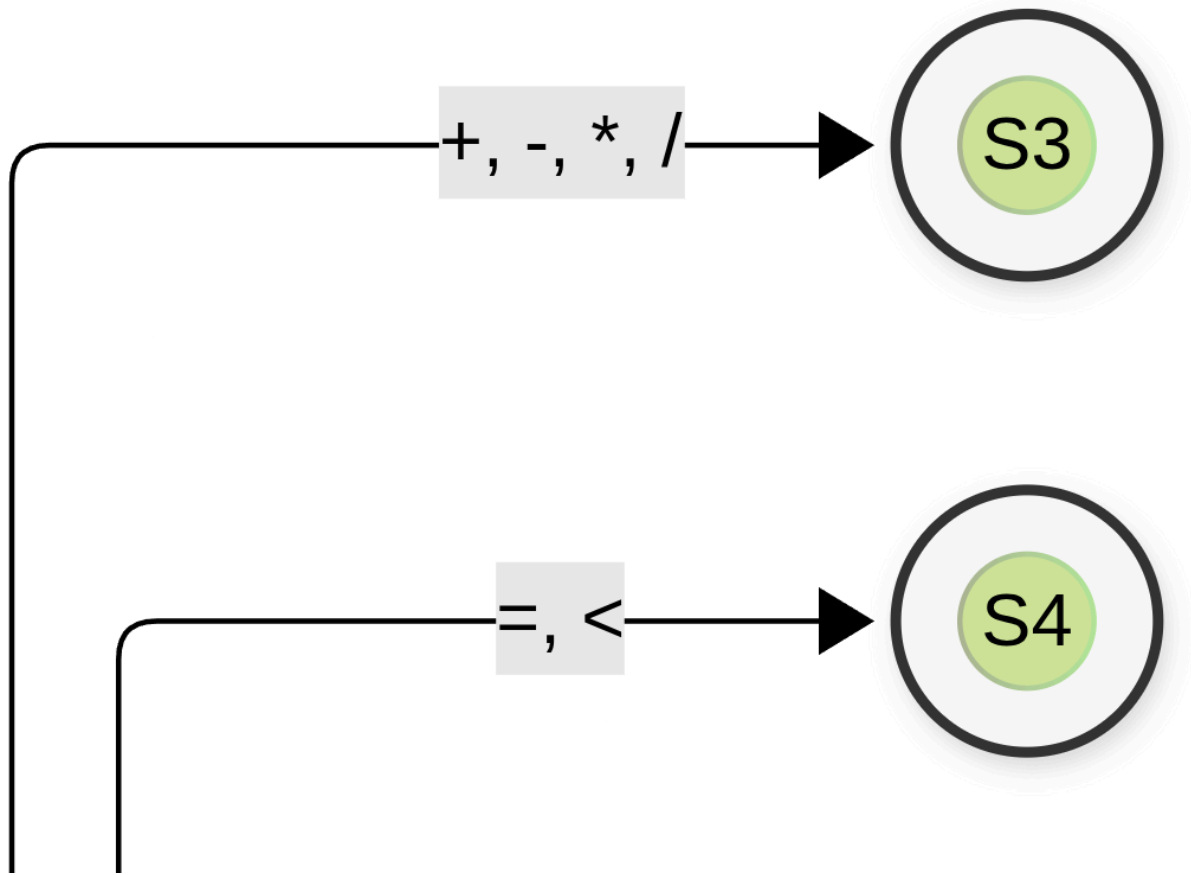
Keywords are recognized via identifier DFA + lookup:

- if
- then
- else
- end
- repeat
- until
- read
- write

- ✓ Same pattern as Identifier
- ✓ After acceptance → check reserved words table

4. Operators

Arithmetic & Relational Operators



$$(q0) \xrightarrow{+} (q3)(SUMOP)$$

$$(q0) \xrightarrow{-} (q3)(SUBOP)$$

$$(q0) \xrightarrow{*} (q3)(MULOP)$$

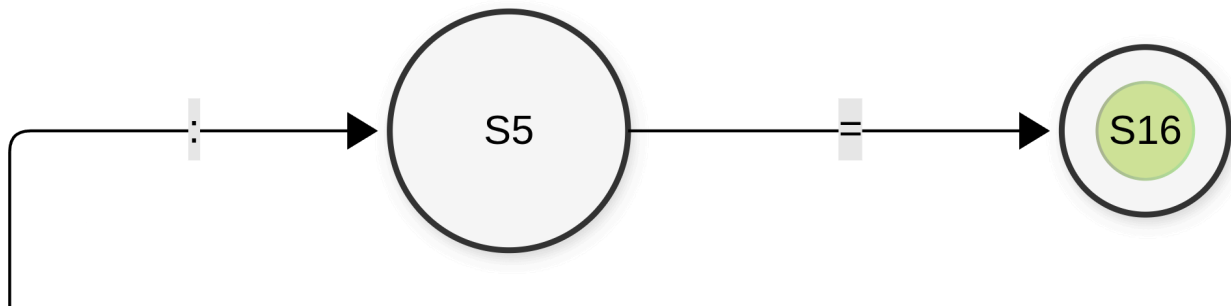
$$(q0) \xrightarrow{/} (q3)(DIVOP)$$

$$(q0) \xrightarrow{=} (q4)$$

$$(q0) \xrightarrow{<} (q4)$$

✓ Accepting state: q3

Assignment Operator

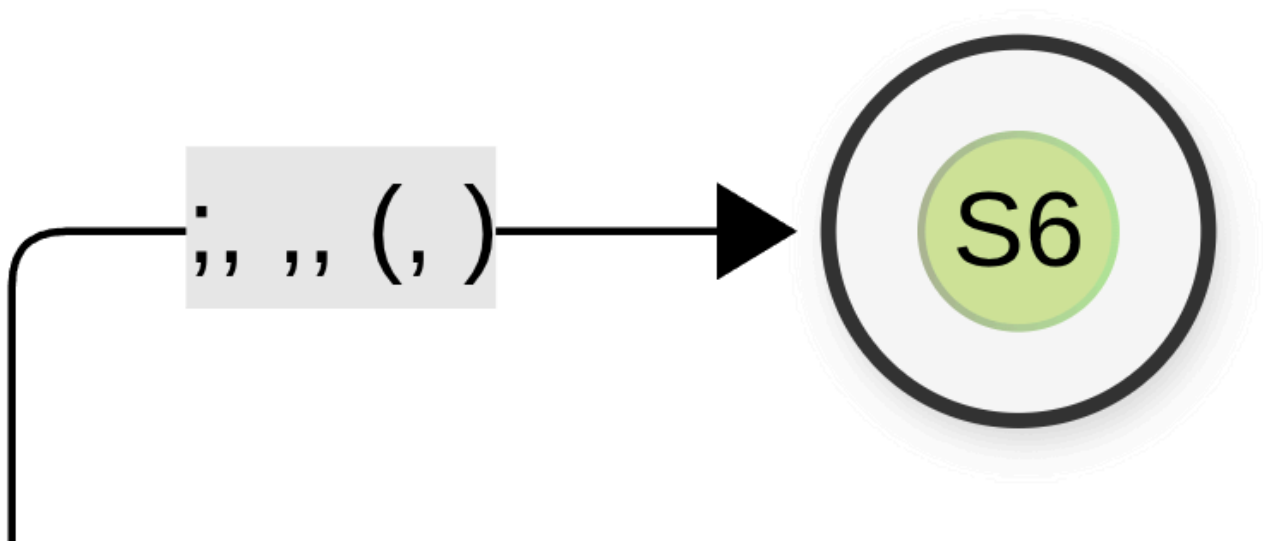


$$(q0) \xrightarrow{\cdot} (q5)$$

$$(q4) \xrightarrow{=} (q16)$$

✓ "[:=" is accepted as Operator

5. Punctuation



$$(q0) \xrightarrow{\cdot} (q6)(SEMICOLON)$$

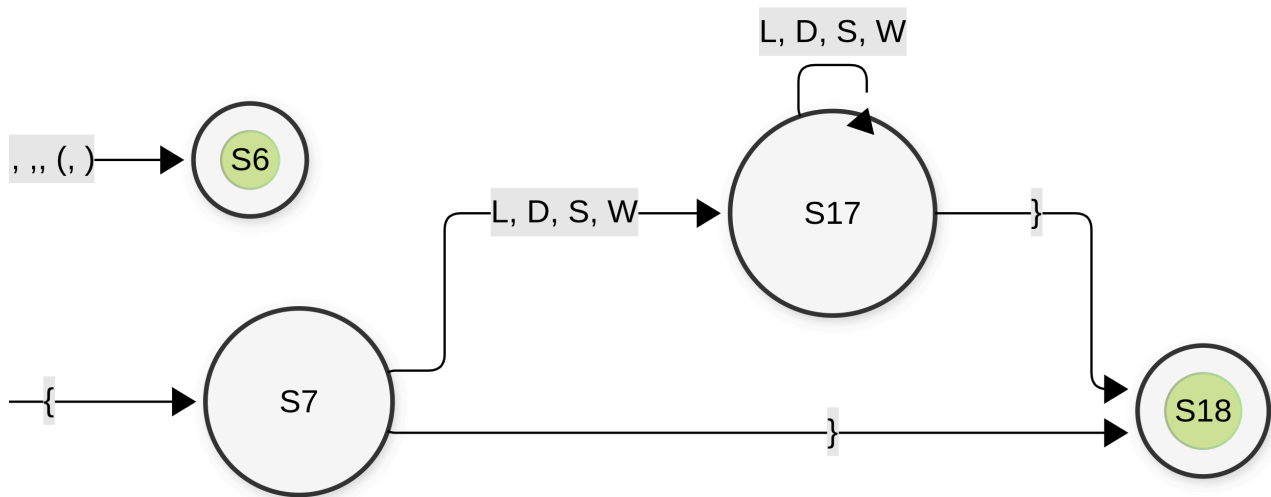
$$(q0) \xrightarrow{\cdot} (q6)(COMMA)$$

$$(q0) \xrightarrow{\text{'('}} (q6)(PUNCTUATION)$$

$$(q0) \xrightarrow{\text{'\)'}} (q6)(PUNCTUATION)$$

✓ Accepting state: q6

6. Comments



Comments are enclosed in `{ }` and cannot be nested.

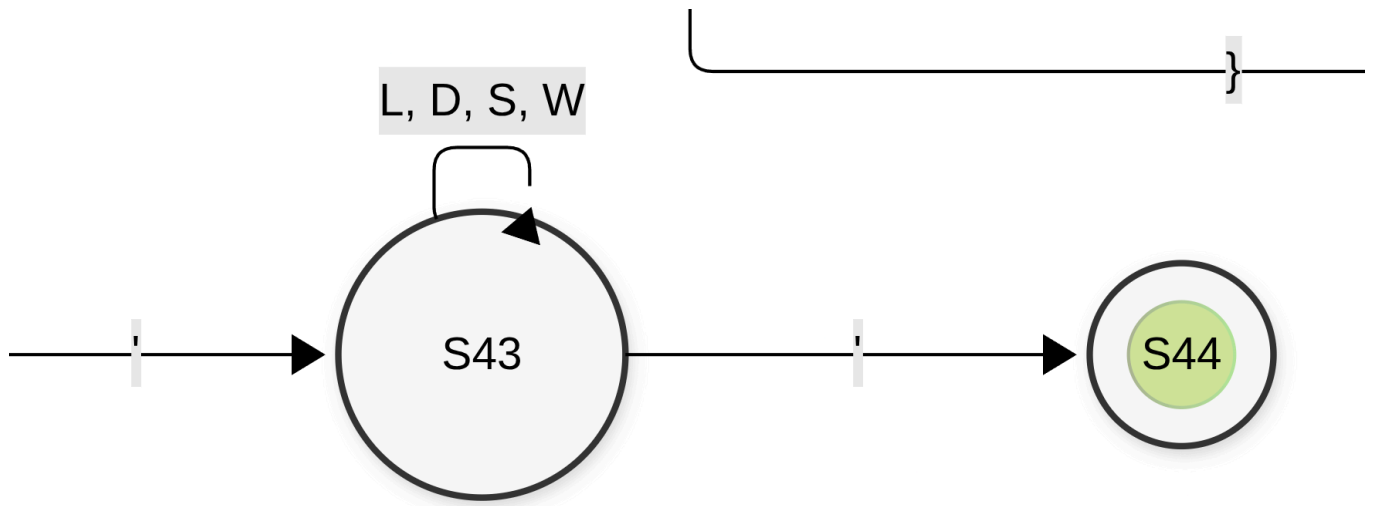
$$(q0) \xrightarrow{\text{'{'}} (q7)$$

$$(q7) \xrightarrow{\text{any except '}'}} (q17)$$

$$(q7) \xrightarrow{\text{'}'}} (q18)$$

✓ Comment is ignored (no token generated)

7. Strings



Strings are enclosed in double quotes.

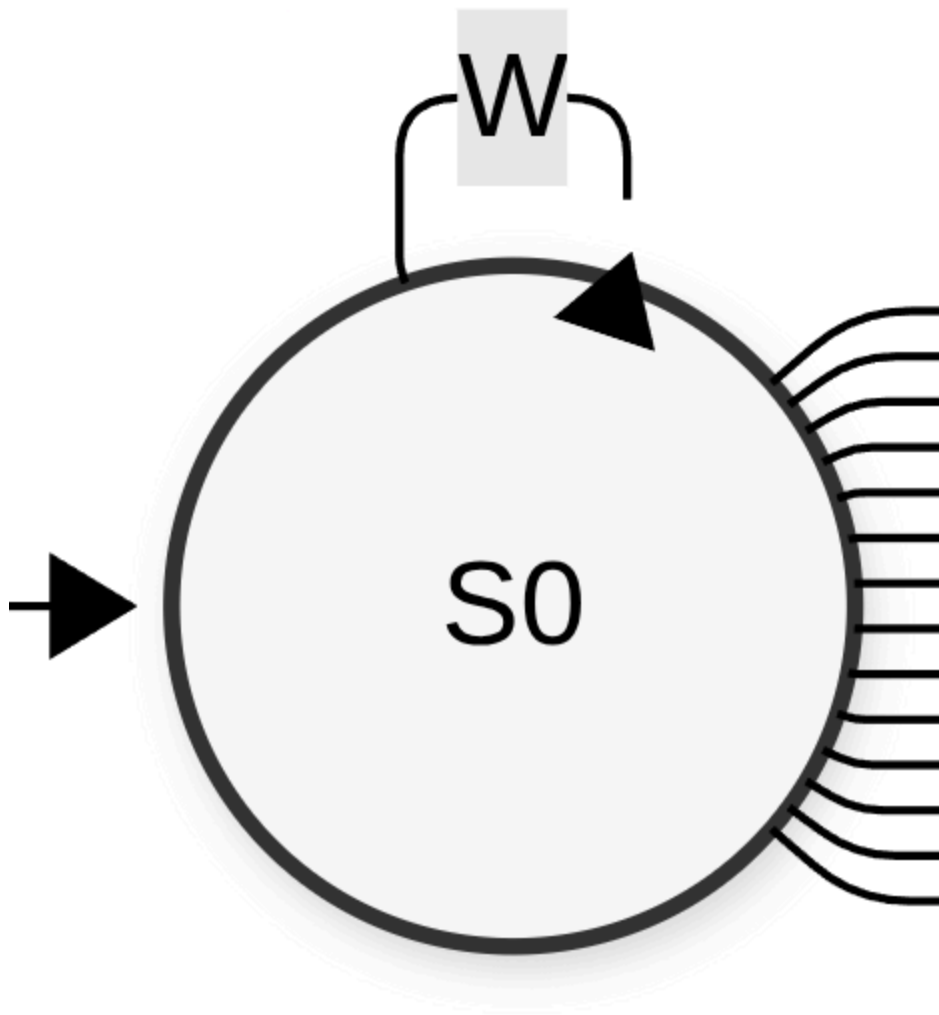
$$(q_0) \xrightarrow{''} (q_{43})$$

$$(q_{43}) \xrightarrow{\text{any except ''}} (q_{43})$$

$$(q_{43}) \xrightarrow{''} (q_{44})$$

✓ Accepting state: q44 (String)

8. Whitespace



Whitespace is ignored.

$$(q_0) \xrightarrow{\text{space/tab/newline}} (q_0)$$

✓ No token generated